

Assignment 2

COM 837 — ML for Wireless Communication Systems

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All results are produced by the code in the accompanying GitHub repository. The Question 1 dataset comes from the provided `generate_dataset.py`, used unmodified; the Question 2 dataset from `generate_qam_dataset.py` with seed 67. All seeds are fixed (`random_state = 42` for splits and K-means), so every number below is reproducible.

1 SVM for LOS/NLOS Identification

1.1 Feature Extraction

Each realisation has six complex taps with delays. The tap amplitudes and powers are

$$|h_l| = \sqrt{h_{\text{real},l}^2 + h_{\text{imag},l}^2}, \quad P_l = |h_l|^2, \quad l = 0, \dots, L-1, \quad L = 6. \quad (1)$$

Five features are extracted per realisation. Each is a statistic over the six taps of one snapshot, so all moments are population statistics (`ddof = 0`).

1.1.1 Kurtosis

$$\mathcal{K} = \frac{\mathbb{E}[(|h| - \mu_{|h|})^4]}{\sigma_{|h|}^4} \quad (2)$$

Peakedness of the amplitude distribution. LOS has one dominant tap, so it is more peaked.

1.1.2 Skewness

$$\mathcal{S} = \frac{\mathbb{E}[(|h| - \mu_{|h|})^3]}{\sigma_{|h|}^3} \quad (3)$$

Asymmetry of the same distribution. Rayleigh (NLOS) and Rician (LOS) differ in asymmetry.

1.1.3 Rising Time

$$\Delta\tau = \tau\left(\arg \max_l |h_l|\right) - \min_l \tau_l \quad (4)$$

Delay gap between the first path and the strongest path. Zero in LOS, where the first path is the strongest.

1.1.4 RMS Delay Spread

$$\tau_m = \frac{\sum_l \tau_l P_l}{\sum_l P_l}, \quad \tau_{\text{rms}} = \sqrt{\frac{\sum_l (\tau_l - \tau_m)^2 P_l}{\sum_l P_l}} \quad (5)$$

Spread of power over delay, in ns. Small in LOS (power concentrated on tap 0), large in NLOS.

1.1.5 Rician K -Factor

$$K_r = \frac{|h|_{\max}^2}{2\sigma_{|h|}^2} \quad (6)$$

Dominant tap power normalised by the variance of the tap amplitudes, exactly as defined in the reference paper. This choice has a consequence, discussed in Section 1.2.1.

1.1.6 Sanity Check

Table 1: Class-conditional feature means at SNR = 30 dB.

Feature	NLOS (−1)	LOS (+1)
Kurtosis	2.16	3.87
Skewness	0.35	1.61
Rising time (ns)	28.15	0.00
RMS delay spread (ns)	45.19	6.98
Rician K -factor, Eq. (6)	6.56	4.58

Four features order themselves as expected. The K -factor of Eq. (6) comes out *lower* for LOS than NLOS; this is a property of the definition, explained in Section 1.2.1.

1.2 Per-Feature and Combined SVM Classification

At each SNR the data are split 80/20 with stratification, a `StandardScaler` is fitted on the training set only, and an RBF-kernel SVM with $C = 1$ is trained. SVM-1 to SVM-5 use one feature each, SVM-6 uses all five. All six share the same split at a given SNR.

Table 2: LOS/NLOS classification accuracy (%) for the six SVM classifiers.

SNR (dB)	Kurtosis	Skewness	Rise time	RMS delay	Rician K	All five
0	59.50	58.75	79.75	88.50	60.00	89.25
5	82.25	84.50	83.00	90.00	68.00	98.25
10	90.75	93.50	81.25	93.00	76.00	99.50
15	93.25	95.00	83.00	97.25	81.75	99.75
20	93.75	94.50	83.00	98.25	81.75	100.00
25	93.50	93.75	82.50	98.25	82.50	100.00
30	93.75	95.50	82.50	98.75	81.50	100.00

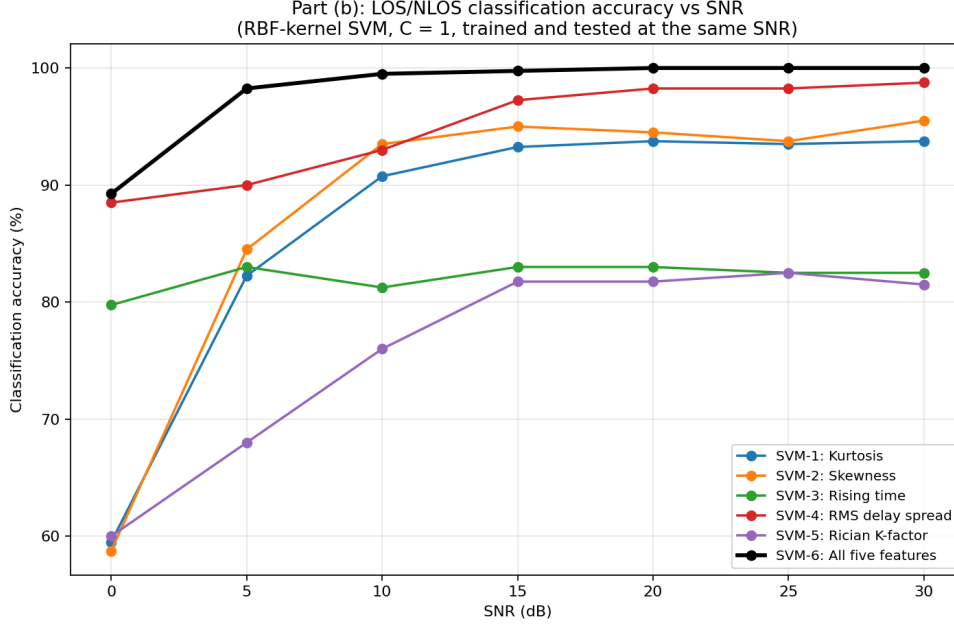


Figure 1: Classification accuracy versus SNR for the six SVM classifiers.

1.2.1 Discussion

Which single feature performs best, and why? RMS delay spread, at every SNR (88.50% at 0 dB to 98.75% at 30 dB). The generator draws LOS delays from an exponential of mean 30 ns and NLOS from one of mean 100 ns, so the classes differ both in how power is spread over the taps and in the span of delays those taps occupy. τ_{rms} captures both at once, giving class means of 45.19 ns against 6.98 ns. The other four features use only the amplitude distribution and discard the delay axis.

Does combining all five features help, and where most? Yes, at every SNR. The gain of SVM-6 over the best single feature is +0.75, +8.25, +6.50, +2.50, +1.75, +1.75 and +1.25 percentage points at 0–30 dB.

It is largest at 5 and 10 dB, where the delay-domain feature is already useful and the amplitude-domain features have just become usable; the two families make different mistakes, so their errors are weakly correlated. At 0 dB the other features are too noisy to contribute, and above 20 dB τ_{rms} alone is already at 98%, leaving no headroom.

Which feature fails at low SNR despite being physically meaningful? The Rician K -factor: 60.00% at 0 dB, the weakest of the five. Two mechanisms combine.

Low-SNR failure. The noise variance is $1/(L \cdot \text{SNR}) = 0.167$ per tap at 0 dB, while the five scattered taps of a LOS realisation carry only about 0.02 each. The noise is roughly eight times stronger than the taps it sits on, so those taps report the noise floor instead of the channel and the LOS statistics migrate onto the NLOS ones (Table 3). This hits kurtosis, skewness and the K -factor alike, which is why all three sit at 58–60% at 0 dB. Only τ_{rms} survives, since its separation rests on the noiseless τ columns.

Table 3: Feature drift with SNR, as LOS mean $\rightarrow$ NLOS mean.

Feature	30 dB	10 dB	0 dB
Kurtosis	3.87 $\rightarrow$ 2.16	3.73 $\rightarrow$ 2.19	2.55 $\rightarrow$ 2.17
Skewness	1.61 $\rightarrow$ 0.35	1.53 $\rightarrow$ 0.39	0.69 $\rightarrow$ 0.30
Rician K	4.58 $\rightarrow$ 6.56	5.06 $\rightarrow$ 7.09	6.97 $\rightarrow$ 8.10
RMS delay spread (ns)	6.98 $\rightarrow$ 45.19	11.91 $\rightarrow$ 51.89	20.47 $\rightarrow$ 65.37

A second, SNR-independent problem. Eq. (6) divides by the variance of the tap amplitudes. With $L = 6$ and one dominant tap of amplitude a , $\sigma_{|h|}^2 \approx a^2(L-1)/L^2$, so $K_r \approx L^2/[2(L-1)] = 3.6$ however strong the LOS component is. The estimator saturates, which is why the LOS mean falls below the NLOS mean and the feature plateaus near 82% even at 30 dB. Repeating the experiment with the textbook form $K = P_{\text{peak}}/(P_{\text{total}} - P_{\text{peak}})$ confirms this is the definition and not the physics:

Table 4: Single-feature accuracy (%) under the two K -factor definitions.

SNR (dB)	0	5	10	15	20	25	30
Eq. (6)	60.00	68.00	76.00	81.75	81.75	82.50	81.50
$P_{\text{peak}}/P_{\text{residual}}$	65.75	87.75	96.25	96.75	97.50	98.00	98.00

The standard form reaches 98% at high SNR while still collapsing to 65.75% at 0 dB. The main results retain Eq. (6) as specified.

1.3 Training Strategy Comparison

SVM-6 is trained once on the 25 dB training split and evaluated on every SNR's test split. The scaler is frozen with the classifier, since a deployed receiver would not know the target-SNR statistics. Both curves use identical test sets.

Table 5: Matched-SNR training versus training only at 25 dB, accuracy (%).

SNR (dB)	Train = Test SNR	Train at 25 dB	Gap (pp)
0	89.25	58.00	− 31.25
5	98.25	78.25	− 20.00
10	99.50	97.50	−2.00
15	99.75	99.50	−0.25
20	100.00	99.75	−0.25
25	100.00	100.00	0.00
30	100.00	100.00	0.00

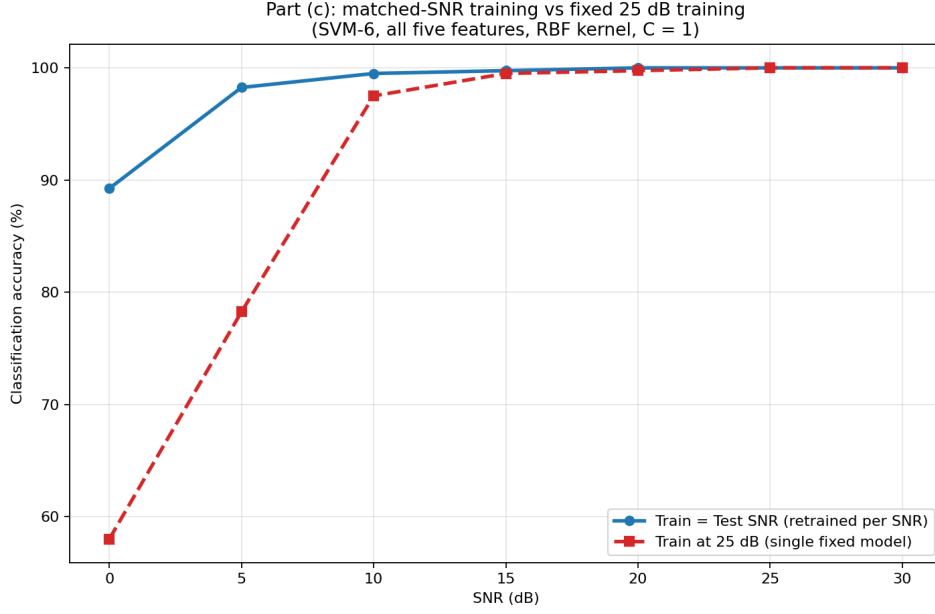


Figure 2: Matched-SNR training versus a single model trained only at 25 dB.

1.3.1 Discussion

At high SNR, do the two curves agree? Yes — identical at 25 and 30 dB and within 0.25 pp at 15 and 20 dB. Above roughly 12 dB the features are effectively SNR-invariant: from Table 3, the LOS kurtosis moves only 3.83 to 3.87 between 15 and 30 dB and the rising time stays at zero. The same region of feature space is occupied at both SNRs, so a boundary fitted at 25 dB is already correct and retraining has nothing to adapt to.

At low SNR, which strategy is better, and by how much? Matched training, by +31.25 pp at 0 dB and +20.00 pp at 5 dB. At 0 dB the fixed model is only 8 pp above a coin flip.

The cause is covariate shift. The 25 dB model learned a boundary for clusters at (kurtosis 3.87, τ_{rms} 7.2 ns) and (2.16, 45.5 ns); at 0 dB those clusters have moved to (2.55, 20.5 ns) and (2.17, 65.4 ns), so LOS test points land where the model expects NLOS. The frozen scaler compounds this, normalising 0 dB features by 25 dB statistics so they arrive outside the region where the local RBF kernel has support.

Would training at 0 dB and testing across all SNR differ? Yes, and not symmetrically. As SNR rises, feature values move monotonically *inward*: the broad, overlapping 0 dB clusters contract, and the tight high-SNR clusters sit nested inside that overlap at its two extremes. A boundary drawn through the middle of the smeared 0 dB distributions therefore still falls between the high-SNR clusters — roughly right, but hedged against noise that is no longer there. In the other direction there is no such protection, since the 25 dB boundary sits at one extreme and the test points migrate past it.

A 0 dB model should therefore degrade gracefully rather than collapse, while never reaching the 100% that matched training attains. Running it confirms this:

Table 6: Accuracy (%) of SVM-6 trained at one fixed SNR and tested across all SNRs.

Trained at	0 dB	5 dB	10 dB	15 dB	20 dB	25 dB	30 dB
0 dB	89.25	92.25	90.00	90.50	91.25	90.75	89.75
5 dB	80.75	98.25	98.00	97.75	98.25	97.75	97.75
25 dB	58.00	78.25	97.50	99.50	99.75	100.00	100.00

The 0 dB model is flat at about 90% everywhere: robust, but capped. Training at the low end of the operating range buys robustness at the cost of peak accuracy; training at the high end does the reverse.

2 Constellation Demodulation and Clustering using K-Means

2.1 Dataset Generation

The square 16-QAM grid uses coordinates from $\{-3, -1, 1, 3\}$. Its average symbol energy is $\mathbb{E}[I^2] + \mathbb{E}[Q^2] = 10$, so every point is divided by $\sqrt{10}$ to give $E_s = 1$:

$$x_I, x_Q \in \left\{ \frac{-3}{\sqrt{10}}, \frac{-1}{\sqrt{10}}, \frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \right\}. \quad (7)$$

With $E_s = 1$ the noise power is $N_0 = 1/\text{SNR}_{\text{lin}}$, split equally between I and Q . For each SNR in $\{0, 5, \dots, 30\}$ dB, 200 samples per constellation point are generated, giving 3200 rows per SNR and 22,400 in total. Seed 67.

One number is used repeatedly below: after scaling, the nearest-neighbour symbol spacing is $2/\sqrt{10} = 0.6325$.

Table 7: Per-axis noise standard deviation against the symbol spacing 0.6325.

SNR (dB)	0	5	10	15	20	25	30
σ per axis	0.707	0.398	0.224	0.126	0.071	0.040	0.022
σ / spacing	1.12	0.63	0.35	0.20	0.11	0.06	0.04

At 0 dB the noise standard deviation exceeds the distance between adjacent symbols, which explains most of what follows.

2.2 Amplitude and Phase Features

For each received sample $z = x_I + jx_Q$:

$$r = \sqrt{x_I^2 + x_Q^2}, \quad \theta = \text{atan2}(x_Q, x_I) \text{ [rad]}. \quad (8)$$

The two-argument form is used because plain $\arctan(x_Q/x_I)$ returns values in $(-\pi/2, \pi/2)$ and so maps a point and its negation to the same phase, aliasing 8 of the 16 symbols. The cost is measurable: polar-feature purity at 25 dB is 1.0000 with atan2 against 0.5188 with plain $\arctan$.

2.3 K-Means Clustering at SNR = 25 dB

K-means is fitted on the Cartesian features for $K = 2, \dots, 20$ with `k-means++` initialisation, $n_{\text{init}} = 10$ and `random.state = 42`. Inertia is $J = \sum_k \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$, which falls monotonically with K and is read for its elbow. The silhouette coefficient is $s(i) = [b(i) - a(i)] / \max(a(i), b(i))$, which has a genuine maximum.

Table 8: Inertia and silhouette coefficient around the elbow, SNR = 25 dB.

K	12	13	14	15	16	17	18	20
Inertia	167.58	129.00	89.63	49.39	10.22	9.97	9.72	9.24
Silhouette	0.660	0.711	0.769	0.823	0.880	0.847	0.813	0.745

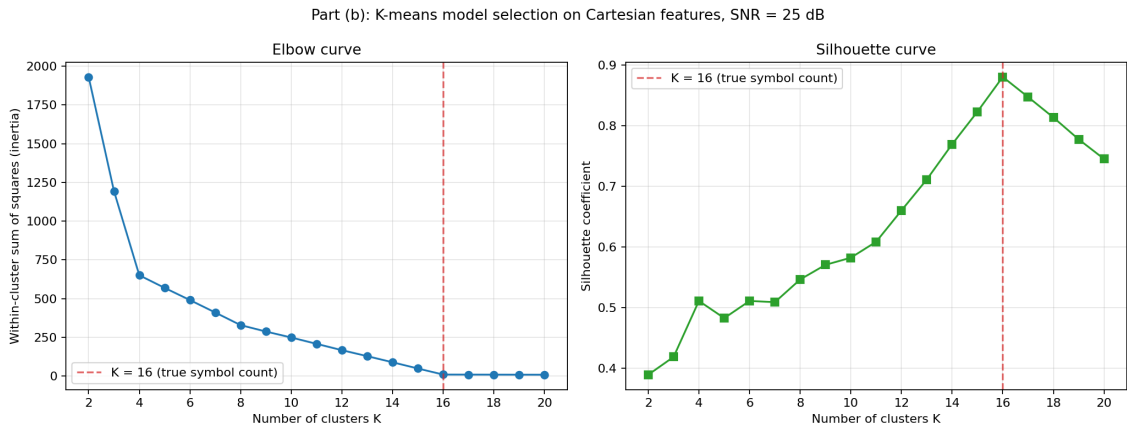


Figure 3: Inertia (left) and silhouette coefficient (right) versus K at SNR = 25 dB. The dashed line marks $K = 16$.

Both metrics select $K = 16$. The inertia drops off a cliff there — 49.39 to 10.22, a 79% reduction in one step — then goes flat. Below 16, at least one cluster must straddle two symbol blobs and the inertia is dominated by that inter-symbol distance; at 16 every cluster covers exactly one blob and only the AWGN variance remains. The silhouette peaks at $K = 16$ ($s = 0.880$) and falls on both sides, with smaller local bumps at $K = 4$ and $K = 8$ corresponding to the quadrants and half-rows of the grid.

2.3.1 Clustering with $K = 16$

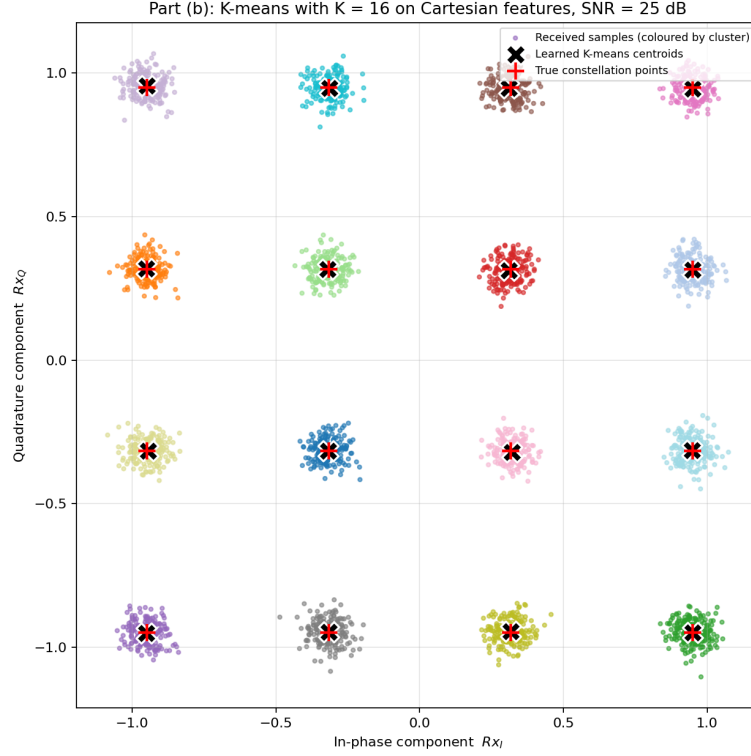


Figure 4: K-means with $K = 16$ on Cartesian features at SNR = 25 dB. Black crosses are the learned centroids, red markers the true constellation points.

The centroids land on the true constellation points with an RMS displacement of 0.0039 against a spacing of 0.6325, and cluster purity is 1.0000. At 25 dB, unsupervised K-means recovers the 16-QAM alphabet perfectly.

2.3.2 Comparison of Feature Sets

Cluster purity assigns each cluster the symbol most frequent within it:

$$\text{Purity} = \frac{1}{N} \sum_k \max_j |C_k \cap G_j|. \quad (9)$$

Three representations are compared at $K = 16$: Set 1 (x_I, x_Q), Set 2 (r, θ) and Set 3 (x_I, x_Q, r, θ) with `StandardScaler` applied. Only Set 3 is scaled, as specified.

Table 9: Cluster purity of the three feature sets across SNR, $K = 16$.

SNR (dB)	Set 1 (Cartesian)	Set 2 (Polar)	Set 3 (Combined, scaled)
0	0.2547	0.2469	0.2431
5	0.4491	0.4150	0.4113
10	0.7656	0.6966	0.6628
15	0.9841	0.9700	0.9678
20	0.9997	0.9994	0.9997
25	1.0000	1.0000	1.0000
30	1.0000	1.0000	1.0000

Which feature set gives the highest purity? At the required 25 dB all three reach 1.0000, so the comparison is saturated: the clusters are about sixteen noise standard deviations apart and even a distorted metric separates them. Across the rest of the range, Set 1 (Cartesian) is highest everywhere, by up to 10 pp at 10 dB. Set 3 finishes last because r and θ are a deterministic function of (x_I, x_Q) and so add no information, only weight: after standardisation the two distorted polar axes influence the distance metric as much as the two clean Cartesian ones.

Why does Euclidean distance suit Cartesian but distort unweighted polar coordinates? The channel adds independent Gaussian noise of equal variance to x_I and x_Q , so its density contours are circles in the I/Q plane and the ML detector is the minimum-Euclidean-distance detector. K-means assumes exactly that — isotropic clusters partitioned by Euclidean Voronoi boundaries — so its bias matches the channel, and since the grid is uniformly spaced, equal distance means equal error probability. Polar coordinates break this in three ways:

1. **Incompatible units.** r spans $[0, 1.34]$ and θ spans $[-\pi, \pi]$, so in $(\Delta r)^2 + (\Delta \theta)^2$ the phase axis dominates and amplitude is under-used.
2. **$\Delta \theta$ is not a distance.** Two points at radius r separated by $\Delta \theta$ are physically $2r \sin(\Delta \theta/2)$ apart, but the metric charges $\Delta \theta$ regardless of r . The inner symbols sit at $r = 0.447$ and the corners at $r = 1.342$, so angular separation is exaggerated threefold near the origin.
3. **θ wraps.** $+\pi$ and $-\pi$ are the same direction but maximally distant in Euclidean terms, so a cluster straddling the negative real axis is split.

The result is wedge- and annulus-shaped decision regions instead of the square Voronoi cells the channel calls for.

2.4 Adaptive K-Means versus Fixed Template

The adaptive strategy fits a fresh $K = 16$ model on each SNR's Cartesian data. The fixed template fits once at 25 dB, freezes the centroids, and only predicts at every SNR.

Table 10: Cluster purity of the two demodulation strategies, $K = 16$.

SNR (dB)	Adaptive K-Means	Fixed Template	Fixed – Adaptive
0	0.2547	0.2631	+0.0084
5	0.4491	0.4722	+ 0.0231
10	0.7656	0.7725	+0.0069
15	0.9841	0.9838	−0.0003
20	0.9997	0.9997	0.0000
25	1.0000	1.0000	0.0000
30	1.0000	1.0000	0.0000

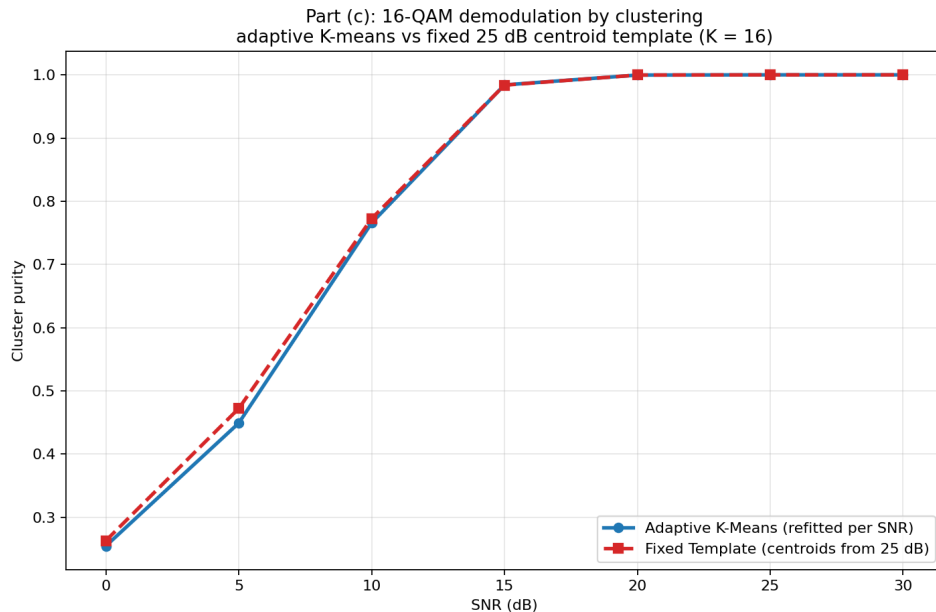


Figure 5: Cluster purity versus SNR for adaptive K-means and the fixed 25 dB template.

2.4.1 Discussion

At high SNR, do the two curves agree? Yes — 1.0000 for both at 25 and 30 dB, 0.9997 at 20 dB, indistinguishable above about 15 dB. The centroids are stable and identifiable there: Table 11 shows the adaptive fit recovering the true grid to within 1–2% of the symbol spacing, so refitting converges to the same sixteen points the template already holds.

Table 11: Adaptive centroid geometry against the true grid (spacing 0.6325). The template’s values are 0.0039 and 0.6258 at all SNRs.

SNR (dB)	RMS error to true grid	Minimum centroid gap
0	0.5744	0.8561
5	0.2347	0.6555
10	0.0536	0.5680
15	0.0112	0.6046
20	0.0062	0.6177
25	0.0039	0.6258
30	0.0021	0.6285

At low SNR, which strategy is more accurate? The fixed template, at every SNR below 15 dB: +0.0231 at 5 dB, +0.0084 at 0 dB. The margin is small, and at 0 dB both are near-useless (0.25 and 0.26 against a chance floor of $1/16$). The template does not rescue the information; it simply never makes matters worse, whereas the adaptive fit sometimes does.

Why does fitting on low-SNR samples cause centroid merging? At 0 dB the per-axis $\sigma = 0.707$ exceeds the symbol spacing of 0.6325, so each Gaussian blob is wider than the gap to its neighbour and all sixteen superpose into one broad cloud. The density minima between symbols are gone, leaving nothing for the algorithm to lock onto. K-means minimises within-cluster sum of squares, so it does the only thing its objective rewards: it tiles that cloud into sixteen roughly equal-mass Voronoi cells, placed to balance point mass rather than to sit on the true symbols.

Table 11 shows the result — the centroids drift an RMS 0.5744 off the grid, 91% of a full symbol spacing, and spread outward until their minimum gap (0.8561) exceeds the true grid's. Covering the wide noise tails leaves the crowded interior under-served, so several true symbols share one cluster and the majority vote inside purity counts the rest as errors.

The fixed template is immune because it is never fitted to the noisy data. Its centroids stay on the true grid, so prediction reduces to a minimum-distance detector — the ML detector for AWGN. It cannot recover what the noise destroyed, but it is optimal given the noise, and so acts as a floor the adaptive fit can match but not beat.

3 Conclusion

Both questions turn on how much structure a learning algorithm should estimate from noisy data rather than be given in advance.

In Question 1 the features split into two families. RMS delay spread reads the noiseless delay axis and is the strongest single feature at every SNR; kurtosis, skewness and the K -factor read only amplitudes across six taps and collapse to 58–60% at 0 dB, where the added noise is about eight times stronger than the scattered taps. Combining all five reaches 100% above 20 dB and gains most at 5–10 dB. Training at a single SNR is safe above roughly 15 dB and costs 31.25 pp at 0 dB, and the failure is one-sided: a 0 dB-trained model instead holds about 90% everywhere without ever reaching 100%.

In Question 2 both the elbow and the silhouette identify $K = 16$, and at 25 dB K-means recovers the constellation exactly. Cartesian coordinates are the right representation because Euclidean distance there is the geometry the AWGN channel imposes. Below 15 dB the fixed template beats adaptive refitting, because once the noise exceeds the symbol spacing K-means tiles the merged cloud by point mass and drifts its centroids almost a full symbol off the grid, while the template stays pinned to the constellation.

In both cases, when the data are clean, estimating and imposing agree. When the data are noisy, estimating a structure that is already known injects error rather than saving effort.